

MH2100 Calculus III

Midterm Exam 2 (Solutions)

AY 2024/2025, Semester 1

Quantitative Research Society @NTU

October 24, 2024

Overview

- **Coverage:** Lagrange multipliers; extrema on disks; and double integrals.
- **Historical source:** MH2100 Midterm Exam 2, AY2024/2025 Semester 1. Every problem below is transcribed from the official paper. Solutions and marking schemes are unofficial QRS@NTU material.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20 (110 minutes), venue TBA. One double-sided typed A4 sheet is permitted; attachments and sticky notes are not.
- **2026 scope note:** spherical coordinates, general Jacobian transformations, and line integrals are beyond the current midterm scope. They are retained and labelled for historical coverage.

Instructions to candidates (historical, verbatim)

- (1) Do not turn over the exam paper until instructed to do so.
- (2) This midterm exam consists of only three problems, each consisting of two parts.
- (3) Candidates are required to attempt all three problems.
- (4) The point allocation for each problem is shown at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) A single double-sided A4-sized help-sheet with only handwritten notes is permitted.

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Question 1

(20 Points)

Problem 1

- (a) Define a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$f(x, y, z) = y^2 - 10z.$$

Let S denote the sphere with center $(0, 0, 0)$ and radius 6, i.e.,

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 36\}.$$

[10 points] Use the Method of Lagrange Multipliers to find the absolute extrema of $f|_S$.

Remark: $f|_S$ denotes the restriction of f to S .

- (b) Let D denote the closed disk in $\mathbb{R}^2$ with center $(0, 0)$ and radius 1, i.e.,

$$D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Define a function $f : D \rightarrow \mathbb{R}$ by

$$f(x, y) = 2x^2 + y^2 - y + 3.$$

- (i) [4 points] Find all the critical points of f , as well as the values of f at these points.

Remark: Any critical point of f must belong to the interior of D .

- (ii) [4 points] Using the standard parametrization $\mathbf{r}$ of the unit circle, given by

$$\forall t \in [0, 2\pi] : \quad \mathbf{r}(t) = (\cos(t), \sin(t)),$$

find the absolute extrema of f on the boundary of D (which is the unit circle).

- (iii) [2 points] Use your work for (b)(i) and (b)(ii) to identify the absolute extrema of f .

Solution

Method 1: Direct calculation

- (a) EVT: S compact, f continuous. Constraint $g = x^2 + y^2 + z^2 = 36$, $\nabla g = 2\langle x, y, z \rangle \neq 0$ on S .

$$\begin{aligned} \nabla f = \lambda \nabla g : \quad & \langle 0, 2y, -10 \rangle = \lambda \langle 2x, 2y, 2z \rangle \\ & -10 = 2\lambda z \implies \lambda \neq 0 & (3\text{rd}) \\ & 0 = 2\lambda x \implies x = 0 & (1\text{st}) \\ & 2y = 2\lambda y \implies y = 0 \text{ or } \lambda = 1 & (2\text{nd}) \\ & y = 0 : z = \pm 6 \implies f = \mp 60 & (g = 36) \\ & \lambda = 1 : z = -5, y = \pm\sqrt{11} \implies f = 61 & (g = 36) \end{aligned}$$

$$\therefore \boxed{\max_S f = 61 \text{ at } (0, \sqrt{11}, -5), (0, -\sqrt{11}, -5); \quad \min_S f = -60 \text{ at } (0, 0, 6).}$$

- (b) (i) $\nabla f = \langle 4x, 2y - 1 \rangle = \mathbf{0} \implies (0, \frac{1}{2}) \in \text{int } D$,

$$\boxed{f(0, 1/2) = 11/4.}$$

(ii) $f(\mathbf{r}(t)) = 2\cos^2 t + \sin^2 t - \sin t + 3$, so with $s = \sin t \in [-1, 1]$:

$$\begin{aligned} f &= 5 - s^2 - s, \quad s \in [-1, 1] & (\cos^2 t = 1 - s^2) \\ f(-\tfrac{1}{2}) &= \tfrac{21}{4}, \quad f(1) = 3, \quad f(-1) = 5 & (\text{vertex; endpoints}) \end{aligned}$$

$$\therefore \boxed{\max_{\partial D} f = 21/4 \text{ at } (\sqrt{3}/2, -1/2), (-\sqrt{3}/2, -1/2); \quad \min_{\partial D} f = 3 \text{ at } (0, 1).}$$

(iii) Candidates $\{11/4, 21/4, 3\}$:

$$\therefore \boxed{\min_D f = 11/4 \text{ at } (0, 1/2); \quad \max_D f = 21/4 \text{ at } (\sqrt{3}/2, -1/2), (-\sqrt{3}/2, -1/2).}$$

Method 2: completing the square

(a)

$$\begin{aligned} f &= 61 - x^2 - (z + 5)^2 \leq 61 & (y^2 = 36 - x^2 - z^2; = \text{ at } (0, \pm\sqrt{11}, -5)) \\ f &\geq -10z \geq -60 & (y^2 \geq 0, z \leq 6; = \text{ at } (0, 0, 6)) \end{aligned}$$

(b)

$$\begin{aligned} f &= 2x^2 + (y - \tfrac{1}{2})^2 + \tfrac{11}{4} \geq \tfrac{11}{4} & (= \text{ at } (0, \tfrac{1}{2})) \\ f &\leq 2(1 - y^2) + y^2 - y + 3 = \tfrac{21}{4} - (y + \tfrac{1}{2})^2 \leq \tfrac{21}{4} & (x^2 \leq 1 - y^2; = \text{ at } (\pm\tfrac{\sqrt{3}}{2}, -\tfrac{1}{2})) \end{aligned}$$

UNOFFICIAL QRS marking scheme (20 points)

(a) **(10 points)** 2: invokes EVT; 3: Lagrange equations and branches; 3: solves all branches; 2: correct extrema and points.

(b) **(10 points)**

(i) **(4 points)** 2: critical equations; 1: point; 1: value.

(ii) **(4 points)** 2: boundary parametrisation; 1: extrema; 1: points.

(iii) **(2 points)** compares all candidates and states global extrema.

Question 2

(15 Points)

Problem 2

(a) [5 points] Evaluate

$$\int_0^1 \left[\int_{\sqrt{y}}^1 \frac{1}{\sqrt{x^3+1}} dx \right] dy.$$

Remark: First sketch the region of integration.(b) (*beyond 2026 midterm scope*) Let E denote the region in the first octant of $\mathbb{R}^3$ (i.e., $\mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$) that is

- bounded above by the sphere given by the equation $x^2 + y^2 + z^2 = 1$;
- bounded below by the closed cone given by the equation $z = \sqrt{\frac{x^2 + y^2}{3}}$.

(i) [3 points] Describe E in spherical coordinates, using inequalities for ρ , θ , and ϕ .(ii) [7 points] Evaluate $\iiint_E 3z \, dV(x, y, z)$ using your work for (b)(i). **Caution:** Work that does not use spherical coordinates will not receive credit.

Solution

Method 1: Direct calculation

(a) Region: $0 \leq y \leq 1, \sqrt{y} \leq x \leq 1 \iff 0 \leq x \leq 1, 0 \leq y \leq x^2$.

$$\begin{aligned} \int_0^1 \int_{\sqrt{y}}^1 \frac{dx \, dy}{\sqrt{x^3+1}} &= \int_0^1 \int_0^{x^2} \frac{dy \, dx}{\sqrt{x^3+1}} && \text{(Fubini)} \\ &= \int_0^1 \frac{x^2}{\sqrt{x^3+1}} dx && (\int dy) \\ &= \frac{1}{3} \int_1^2 w^{-1/2} dw && (w = x^3 + 1) \end{aligned}$$

$$\boxed{\int_0^1 \frac{x^2}{\sqrt{x^3+1}} dx = \frac{2}{3} (\sqrt{2} - 1)}.$$

(b) *Beyond 2026 midterm scope.*

$$\begin{aligned} \text{first octant} &\implies 0 \leq \theta \leq \frac{\pi}{2} \\ z = \sqrt{(x^2 + y^2)/3} &\iff \tan \phi = \sqrt{3} \implies 0 \leq \phi \leq \frac{\pi}{3} && \text{(above cone)} \\ x^2 + y^2 + z^2 \leq 1 &\implies 0 \leq \rho \leq 1 \end{aligned}$$

$$\boxed{0 \leq \rho \leq 1, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{3}.$$

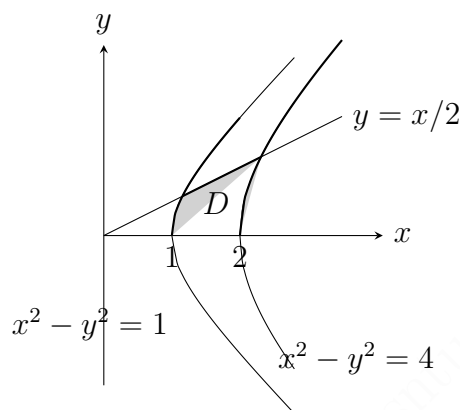
$$\begin{aligned} \iiint_E 3z \, dV &= \int_0^{\pi/2} \int_0^{\pi/3} \int_0^1 3\rho^3 \cos \phi \sin \phi \, d\rho \, d\phi \, d\theta && (z = \rho \cos \phi, \rho^2 \sin \phi) \\ &= \frac{\pi}{2} \cdot \frac{3}{8} \cdot \frac{3}{4} && (\theta, \phi, \rho) \\ &= \boxed{\frac{9\pi}{64}} \end{aligned}$$

UNOFFICIAL QRS marking scheme (15 points)

- (a) **(5 points)** 2: sketches/reverses the region; 2: integrates in y ; 1: substitutes and obtains $\frac{2}{3}(\sqrt{2} - 1)$.
- (b) **(10 points)**
- (i) **(3 points)** 1: each spherical bound.
- (ii) **(7 points)** 2: correct integrand and Jacobian; 3: integral; 2: value $9\pi/64$.

Question 3*(beyond 2026 midterm scope)***(15 Points)****Problem 3**(a) Let D denote the region in the first quadrant of $\mathbb{R}^2$ bounded by the following four curves:

- The two hyperbolas given by the equations $x^2 - y^2 = 1$ and $x^2 - y^2 = 4$.
- The two lines given by the equations $y = 0$ and $y = x/2$.

[7 points] Evaluate $\iint_D \frac{y}{x} dA(x, y)$ using a change of variables.(b) (i) Define a function $f : \mathbb{R}^3 \setminus \{(0, 0, 0)\} \rightarrow \mathbb{R}$ by

$$f(x, y, z) = \frac{1}{x^2 + y^2 + z^2}.$$

Let C denote the curve in $\mathbb{R}^3$ having the parametrization $\mathbf{r} : [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{r}(t) = (\cos t, \sin t, t).$$

[4 points] Evaluate $\int_C f ds$.(ii) Define a vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) = (x, 3xy, -(x + z)).$$

Let C denote the directed line segment going from $(1, 4, 2)$ to $(0, 5, 1)$.[4 points] Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.**Solution****Method 1: Direct calculation**(a) *Beyond 2026 midterm scope.* $u = x^2 - y^2$, $v = y/x$.

$$D \mapsto \{1 \leq u \leq 4, 0 \leq v \leq \tfrac{1}{2}\} \quad (\text{boundaries})$$

$$(u, v) \mapsto (\sqrt{u/(1-v^2)}, vx) = (x, y) \quad (\text{inverse on } x > 0, \text{ one-to-one on } D)$$

$$\frac{\partial(u, v)}{\partial(x, y)} = 2(1 - v^2) \implies dA = \frac{du dv}{2(1 - v^2)}$$

$$\iint_D \frac{y}{x} dA = \int_1^4 \int_0^{1/2} \frac{v}{2(1 - v^2)} dv du$$

$$= 3 \left[-\frac{1}{4} \ln(1 - v^2) \right]_0^{1/2} = \boxed{\frac{3}{4} \ln \left(\frac{4}{3} \right)}$$

(b) *Beyond 2026 midterm scope.*

$$(i) \quad \|\mathbf{r}'(t)\| = \sqrt{2}, \quad f(\mathbf{r}(t)) = \frac{1}{1+t^2}$$

$$\boxed{\int_C f \, ds = \sqrt{2} \int_0^{2\pi} \frac{dt}{1+t^2} = \sqrt{2} \arctan(2\pi).}$$

$$(ii) \quad \mathbf{r}(t) = \langle 1-t, 4+t, 2-t \rangle, \quad 0 \leq t \leq 1:$$

$$\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (1-t, 3(1-t)(4+t), 2t-3) \cdot \langle -1, 1, -1 \rangle = 14 - 10t - 3t^2$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^1 (14 - 10t - 3t^2) \, dt = \boxed{8}$$

UNOFFICIAL QRS marking scheme (15 points)

(a) **(7 points)** 2: valid transformation; 2: maps region; 2: Jacobian; 1: value.

(b) **(8 points)**

(i) **(4 points)** 1: parametrisation; 1: arc length; 1: integral; 1: value.

(ii) **(4 points)** 1: segment parametrisation; 2: dot product; 1: value 8.